

LAPORAN PRAKTIKUM 1
SISTEM OPERASI



Dosen Pembimbing:

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PROGRAM STUDI TEKNIK INFORMATIKA
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1. Operating - System Services

- Menjelajahi Layanan OS

Lihat informasi sistem

```
nadiva_nathania@LAPTOP-JR1G744R:~$ uname -a
lsb_release -a
Linux LAPTOP-JR1G744R 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
```

Lihat proses yang berjalan

```
nadiva_nathania@LAPTOP-JR1G744R:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.1  0.3 21676 12376 ?        Ss   19:47   0:00 /sbin/init
root         2  0.0  0.0   3072 1664 ?        SL   19:47   0:00 /init
root         8  0.0  0.0   3072 1792 ?        SL   19:47   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
--log-truncate
root        44  0.0  0.4 66828 16376 ?        S<s  19:47   0:00 /usr/lib/systemd/systemd-journald
root        91  0.0  0.1 25268 6144 ?        Ss   19:47   0:00 /usr/lib/systemd/systemd-udevd
systemd+  106  0.0  0.3 21456 11776 ?        Ss   19:47   0:00 /usr/lib/systemd/systemd-resolved
systemd+  107  0.0  0.1 91024 7424 ?        Ssl  19:47   0:00 /usr/lib/systemd/systemd-timesyncd
root       156  0.0  0.0  4236 2432 ?        Ss   19:47   0:00 /usr/sbin/cron -f -P
message+  157  0.0  0.1  9624 4480 ?        Ss   19:47   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
stemd-activation --syslog-only
root       164  0.0  0.2 17060 8192 ?        Ss   19:47   0:00 /usr/lib/systemd/systemd-logind
root       166  0.0  0.3 1755840 12160 ?       Ssl  19:47   0:00 /usr/libexec/ssl-pro-service -vv
root       172  0.0  0.0   3160 1920 hvc0    Ss+  19:47   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600
vt220
syslog    186  0.0  0.1 222508 5248 ?        Ssl  19:47   0:00 /usr/sbin/rsyslogd -n -iNONE
root     189  0.0  0.0   3116 1792 tty1    Ss+  19:47   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root     198  0.0  0.5 107012 22016 ?       Ssl  19:47   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade
-shutdown --wait-for-signal
root     285  0.0  0.0   3076 1024 ?        Ss   19:47   0:00 /init
root     286  0.0  0.0   3092 1156 ?        S    19:47   0:00 /init
nadiva_+  287  0.0  0.1  6068 5248 pts/0    Ss   19:47   0:00 -bash
root     288  0.0  0.1  6664 4352 pts/1    Ss   19:47   0:00 /bin/login -f
```

Lihat penggunaan memori

```
nadiva_nathania@LAPTOP-JR1G744R:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.7Gi         442Mi        2.9Gi         3.5Mi         427Mi        3.2Gi
Swap:           1.0Gi           0B         1.0Gi
```

Lihat informasi CPU

```
nadiva_nathania@LAPTOP-JR1G744R:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 12
On-line CPU(s) list: 0-11
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i5-13420H
CPU Family: 6
Model: 186
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 2
BogoMIPS: 5222.39
Flags: fpu_vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht sys
call nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_fre
q pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx
f16c rdrand hypervisor lahf_lm abm 3dnowprefetch sabb ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept
ad fsgbase tsc_adjust bti avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xs
avec xgetbv1 xsaves avx_vnni vnni unpkg waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear s
erialize flush_l1d arch_capabilities

Virtualization features:
Virtualization: VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 288 KiB (6 instances)
L1i: 192 KiB (6 instances)
L2: 7.5 MiB (6 instances)
L3: 12 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-11
Vulnerabilities:
Gather data sampling: Not affected

Gather data sampling: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBSRB-eIBRS SW sequence; BHI BHI_DIS_
S
Srbds: Not affected
Tsx async abort: Not affected
nadiva_nathania@LAPTOP-JR1G744R:~$
```

Lihat perangkat yang terpasang

```
nadiva_nathania@LAPTOP-JR1G744R:~$ lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 388.4M  1 disk
sdb   8:16   0   186M  1 disk
sdc   8:32   0    1G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro
/
```

Praktik ini mempelajari tentang berbagai layanan dasar yang disediakan oleh sistem operasi, seperti layanan eksekusi program, manajemen file, I/O, serta manajemen proses, tujuannya agar memahami peran utama sistem operasi dalam mengelola sumber daya komputer dan menyediakan antarmuka antara perangkat keras dan pengguna.

- **Program C - Menggunakan Layanan OS**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano os_services.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -o os_services os_services.c
./os_services
=== Program Execution Service ===
Process ID (PID): 494
Parent Process ID (PPID): 429

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
nadiva_nathania@LAPTOP-JR1G744R:~$ |
```

Praktik ini menunjukkan bagaimana sistem operasi mengeksekusi suatu program dan memberikan identitas unik pada setiap proses yang berjalan dengan menggunakan fungsi getpid() dan getppid(), mahasiswa jadi mengerti tentang Process ID (PID) dan Parent Process ID (PPID). Praktik ini bertujuan untuk memahami hubungan hierarki antar proses di dalam sistem operasi.

2. User and Operating-System Interface

- **Command Line Interface (CLI)**

Navigasi dasar

```

nadiva_nathanian@LAPTOP-JR1G744R:~$ pwd
/home/nadiva_nathanian
nadiva_nathanian@LAPTOP-JR1G744R:~$ ls -la
total 1296
drwxr-x--- 9 nadiva_nathanian nadiva_nathanian 4096 Oct 16 22:16 .
drwxr-xr-x 3 root root 4096 Oct 10 13:13 ..
-rw----- 1 nadiva_nathanian nadiva_nathanian 16985 Oct 16 20:55 .bash_history
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 220 Oct 10 13:13 .bash_logout
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 3771 Oct 10 13:13 .bashrc
drwx----- 3 nadiva_nathanian nadiva_nathanian 4096 Oct 16 12:33 .cache
drwx----- 3 nadiva_nathanian nadiva_nathanian 4096 Oct 15 20:37 .config
-rw----- 1 nadiva_nathanian nadiva_nathanian 20 Oct 16 21:41 .lessshst
drwxr-xr-x 3 nadiva_nathanian nadiva_nathanian 4096 Oct 10 14:03 .local
-rw-rw-r-- 1 nadiva_nathanian nadiva_nathanian 0 Oct 16 09:04 .motd_shown
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 807 Oct 10 13:13 .profile
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 0 Oct 10 14:14 .sudo_as_admin_successful
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 161 Oct 15 20:23 Makefile
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 512 Oct 16 22:01 bootloader.bin
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 382 Oct 16 12:24 bootloader_concept.asm
-rwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 17792 Oct 16 12:32 buggy
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 344 Oct 16 12:32 buggy.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 61 Oct 11 07:51 file.txt
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16352 Oct 14 21:34 file_syscalls
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 1000 Oct 14 21:29 file_syscalls.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 52 Oct 11 07:28 files.txt
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 439 Oct 15 20:19 hello_module.c
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16384 Oct 14 21:35 info_syscalls
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 712 Oct 14 21:30 info_syscalls.c
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 15144 Oct 15 19:12 libmathops.so
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 148 Oct 15 18:48 main.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 21446 Oct 15 18:59 main.i
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 1728 Oct 15 19:07 main.o
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 910 Oct 15 19:07 main.s
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 113 Oct 15 18:58 math_ops.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 98 Oct 15 18:46 math_ops.h
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 317 Oct 15 19:07 math_ops.i
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 1352 Oct 15 19:12 math_ops.o

-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 98 Oct 15 18:46 math_ops.h
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 317 Oct 15 19:07 math_ops.i
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 1352 Oct 15 19:12 math_ops.o
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 992 Oct 15 19:07 math_ops.s
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 17288 Oct 16 12:37 memleak
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 115 Oct 16 12:37 memleak.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 805 Oct 16 11:21 my_daemon.c
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 21992 Oct 15 14:55 myshell
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 13869 Oct 15 14:46 myshell.c
drwxr-xr-x 2 nadiva_nathanian nadiva_nathanian 4096 Oct 11 08:09 myshell_project
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16488 Oct 14 20:04 os_services
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 1315 Oct 12 23:34 os_services.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 675 Oct 15 20:01 os_specific.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 18 Oct 11 09:54 out.txt
drwxr-xr-x 3 nadiva_nathanian nadiva_nathanian 4096 Oct 10 19:59 praktikum_os
drwxr-xr-x 4 nadiva_nathanian nadiva_nathanian 4096 Oct 10 20:00 praktikum_os
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16360 Oct 14 21:33 process_syscalls
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 873 Oct 14 21:15 process_syscalls.c
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16256 Oct 16 12:04 procfs_read
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 1051 Oct 15 21:15 procfs_read.c
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16056 Oct 15 19:08 program
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16056 Oct 15 19:10 program_dynamic
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16024 Oct 15 19:12 program_shared
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 785400 Oct 15 19:10 program_static
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 16528 Oct 16 22:16 simple_shell
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 1517 Oct 15 18:12 simple_shell.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 25 Oct 14 21:34 syscall_test.txt
drwxr-xr-x 1 nadiva_nathanian nadiva_nathanian 21496 Oct 14 16:33 sysmon
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 13404 Oct 14 16:30 sysmon.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 15455 Oct 14 11:56 sysnom.c
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 257 Oct 15 14:52 test.txt
drwxr-xr-x 2 nadiva_nathanian nadiva_nathanian 4096 Oct 15 18:14 test_dir
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 23 Oct 14 20:04 test_file.txt
-rw-r--r-- 1 nadiva_nathanian nadiva_nathanian 11247 Oct 16 13:27 trace.txt
nadiva_nathanian@LAPTOP-JR1G744R:~$ cd /tmp
nadiva_nathanian@LAPTOP-JR1G744R:/tmp$ mkdir test_dir
nadiva_nathanian@LAPTOP-JR1G744R:/tmp$ cd test_dir

```

Pipelining

```

nadiva_nathanian@LAPTOP-JR1G744R:/tmp/test_dir$ ls -l | grep "^d"
nadiva_nathanian@LAPTOP-JR1G744R:/tmp/test_dir$ ps aux | grep bash | wc -l
3

```

Redirection

```

nadiva_nathanian@LAPTOP-JR1G744R:/tmp/test_dir$ echo "Hello World" > output.txt
nadiva_nathanian@LAPTOP-JR1G744R:/tmp/test_dir$ cat output.txt
Hello World
nadiva_nathanian@LAPTOP-JR1G744R:/tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
nadiva_nathanian@LAPTOP-JR1G744R:/tmp/test_dir$ cat < output.txt
Hello World

```

Background processes

```
nadiva_nathania@LAPTOP-JR1G744R:/tmp/test_dir$ sleep 30 &
[1] 996
nadiva_nathania@LAPTOP-JR1G744R:/tmp/test_dir$ jobs
[1]+  Running                  sleep 30 &
nadiva_nathania@LAPTOP-JR1G744R:/tmp/test_dir$ fg %1
sleep 30
nadiva_nathania@LAPTOP-JR1G744R:/tmp/test_dir$ |
```

Praktik ini bertujuan mempelajari cara sistem operasi mengatur dan mengelola file menggunakan fungsi standar C seperti fopen(), fprintf(), fscanf(), dan remove(). CLI lebih cepat untuk melakukan tugas tertentu, terutama yang berulang atau kompleks. CLI juga memungkinkan pengguna untuk menulis skrip serangkaian perintah yang dijalankan secara otomatis.

- **Program C - Shell Sederhana**

```
nadiva_nathania@LAPTOP-JR1G744R:/tmp/test_dir$ nano simple_shell.c
nadiva_nathania@LAPTOP-JR1G744R:/tmp/test_dir$ gcc -o simple_shell simple_shell.c
nadiva_nathania@LAPTOP-JR1G744R:/tmp/test_dir$ ./simple_shell
myshell> |
```

Program C - Shell Sederhana ini adalah praktik yang bertujuan untuk memahami bagaimana sistem operasi memuat dan menjalankan program. Shell adalah program yang menerima perintah teks dari pengguna, menguraikannya, dan kemudian mengeksekusi perintahnya.

3. System Calls

- **Process Control System Calls**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano process_syscalls.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -o process_syscalls process_syscalls.c
nadiva_nathania@LAPTOP-JR1G744R:~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 712
Parent waiting for child (PID: 713)...
Child PID: 713
Child's Parent PID: 712
Child executing 'ls' command...
total 216
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 61 Oct 11 07:51 file.txt
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 16352 Oct 14 21:31 file_syscalls
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 1000 Oct 14 21:29 file_syscalls.c
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 52 Oct 11 07:28 files.txt
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 16384 Oct 14 21:31 info_syscalls
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 712 Oct 14 21:30 info_syscalls.c
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 21672 Oct 11 11:13 myshell
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 17374 Oct 11 11:12 myshell.c
drwxr-xr-x 2 nadiva_nathania nadiva_nathania 4096 Oct 11 08:09 myshell_project
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 16488 Oct 14 20:04 os_services
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 1315 Oct 12 23:34 os_services.c
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 18 Oct 11 09:54 out.txt
drwxr-xr-x 3 nadiva_nathania nadiva_nathania 4096 Oct 10 19:59 praktikum.os
drwxr-xr-x 4 nadiva_nathania nadiva_nathania 4096 Oct 10 20:00 praktikum_os
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 16360 Oct 14 21:33 process_syscalls
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 873 Oct 14 21:15 process_syscalls.c
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 25 Oct 14 21:32 syscall_test.txt
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 21496 Oct 14 16:33 sysmon
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 13404 Oct 14 16:30 sysmon.c
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 15455 Oct 14 11:56 sysnom.c
-rw-r--r-- 1 nadiva_nathania nadiva_nathania 23 Oct 14 20:04 test_file.txt
Child exited with status: 0
nadiva_nathania@LAPTOP-JR1G744R:~$ |
```

Process Control System Calls ini bertujuan adalah untuk memungkinkan program pengguna berinteraksi dengan kernel sistem operasi guna mengelola dan mengendalikan proses-proses yang sedang berjalan.

- **File Management System Calls**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano file_syscalls.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -o file_syscalls file_syscalls.c
nadiva_nathania@LAPTOP-JR1G744R:~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
nadiva_nathania@LAPTOP-JR1G744R:~$ |
```

File Management System Calls bertujuan untuk antarmuka yang aman, terstruktur, dan standar bagi program pengguna untuk berinteraksi dengan sistem berkas dari sistem operasi. Ini memungkinkan aplikasi untuk mengelola, mengakses, dan memanipulasi data yang tersimpan di perangkat.

- **Information Maintenance System Calls**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano info_syscalls.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -o info_syscalls info_syscalls.c
nadiva_nathania@LAPTOP-JR1G744R:~$ ./info_syscalls
=== Information Maintenance System Calls ===

Process ID: 729
Parent Process ID: 600
User ID: 1000
Group ID: 1000
Current time: Tue Oct 14 21:35:26 2025
Clock ticks since boot: 1718603630
Sleeping for 2 seconds...
Awake!
```

Information Maintenance System Calls bertujuan untuk memungkinkan program pengguna untuk mengakses dan mengatur data internal sistem yang krusial, seperti waktu, tanggal, data proses, dan statistik sistem, yang berada di bawah kendali kernel.

4. System Services

- **System**

Lihat semua service

```
nadiva_nathania@LAPTOP-JR1G744R:~$ systemctl list-units --type=service
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
console-getty.service	loaded	active	running	Console Getty
console-setup.service	loaded	active	exited	Set console font and keymap
cron.service	loaded	active	running	Regular background program processing daemon
dbus.service	loaded	active	running	D-Bus System Message Bus
getty@tty1.service	loaded	active	running	Getty on tty1
keyboard-setup.service	loaded	active	exited	Set the console keyboard layout
kmod-static-nodes.service	loaded	active	exited	Create List of Static Device Nodes
polkit.service	loaded	active	running	Authorization Manager
rsyslog.service	loaded	active	running	System Logging Service
setvtrgb.service	loaded	active	exited	Set console scheme
snappy.service	loaded	active	exited	Wait until snappy is fully seeded
systemd-binfmt.service	loaded	active	exited	Set Up Additional Binary Formats
systemd-journal-flush.service	loaded	active	exited	Flush Journal to Persistent Storage
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-modules-load.service	loaded	active	exited	Load Kernel Modules
systemd-remount-fs.service	loaded	active	exited	Remount Root and Kernel File Systems
systemd-resolved.service	loaded	active	running	Network Name Resolution
systemd-sysctl.service	loaded	active	exited	Apply Kernel Variables
systemd-timesyncd.service	loaded	active	running	Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service	loaded	active	exited	Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service	loaded	active	exited	Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service	loaded	active	exited	Create Volatile Files and Directories
systemd-udev-trigger.service	loaded	active	exited	Coldplug All udev Devices
systemd-udevd.service	loaded	active	running	Rule-based Manager for Device Events and Files
systemd-update-utmp.service	loaded	active	exited	Record System Boot/Shutdown in UTMP
systemd-user-sessions.service	loaded	active	exited	Permit User Sessions
unattended-upgrades.service	loaded	active	running	Unattended Upgrades Shutdown
user-runtime-dir@1000.service	loaded	active	exited	User Runtime Directory /run/user/1000
user@1000.service	loaded	active	running	User Manager for UID 1000
wsl-pro.service	loaded	active	running	Bridge to Ubuntu Pro agent on Windows

Legend: LOAD → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
SUB → The low-level unit activation state, values depend on unit type.

Status service

```
nadiva_nathania@LAPTOP-JR1G744R:~$ systemctl status ssh
```

```
o ssh.service - OpenSSH Secure Shell server
```

```
Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
```

```
Active: inactive (dead)
```

```
TriggeredBy: ● ssh.socket
```

```
Docs: man:sshd(8)
```

```
man:sshd_config(5)
```

```
nadiva_nathania@LAPTOP-JR1G744R:~$
```

Lihat log service

```
nadiva_nathania@LAPTOP-JR1G744R:~$ journalctl -u ssh -n 20
```

```
-- No entries --
```

Lihat service yang berjalan saat boot


```
nadiva_nathania@LAPTOP-JR1G744R:~$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE    PRESET
apparmor.service                        enabled  enabled
apport.service                          enabled  enabled
cloud-config.service                    enabled  enabled
cloud-final.service                      enabled  enabled
cloud-init-local.service                 enabled  enabled
cloud-init.service                       enabled  enabled
console-setup.service                   enabled  enabled
cron.service                            enabled  enabled
dmesg.service                           enabled  enabled
e2scrub_reap.service                     enabled  enabled
getty@.service                           enabled  enabled
keyboard-setup.service                  enabled  enabled
landscape-client.service                 enabled  enabled
networkd-dispatcher.service              enabled  enabled
rsyslog.service                         enabled  enabled
setvtrgb.service                        enabled  enabled
snapd.apparmor.service                  enabled  enabled
snapd.autoimport.service                 enabled  enabled
snapd.core-fixup.service                 enabled  enabled
snapd.recovery-chooser-trigger.service   enabled  enabled
snapd.seeded.service                     enabled  enabled
snapd.service                           enabled  enabled
snapd.system-shutdown.service            enabled  enabled
systemd-pstore.service                   enabled  enabled
systemd-resolved.service                 enabled  enabled
systemd-timesyncd.service                 enabled  enabled
ua-reboot-cmds.service                   enabled  enabled
ubuntu-advantage.service                 enabled  enabled
unattended-upgrades.service              enabled  enabled
wsl-pro.service                          enabled  enabled

30 unit files listed.
```

Menggunakan systemd bertujuan untuk menyediakan sistem inisialisasi dan manajer layanan yang modern dan cepat untuk sistem operasi Linux. Sebagai proses pertama yang dijalankan oleh kernel, systemd bertanggung jawab untuk mengelola seluruh sistem, mulai dari booting hingga shutdown.

- **Membuat Custom System Service**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano my_daemon.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -o my_daemon my_daemon.c
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo nano /etc/systemd/system/my_daemon.service
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo mv my_daemon /usr/local/bin/
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo systemctl daemon-reload
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo systemctl start my_daemon
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo systemctl status my_daemon
● my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
   Active: active (running) since Thu 2025-10-16 21:14:54 WIB; 7s ago
     Process: 747 ExecStart=/usr/local/bin/my_daemon (code=exited, status=0/SUCCESS)
    Main PID: 748 (my_daemon)
       Tasks: 1 (limit: 4480)
      Memory: 200.0K (peak: 824.0K)
         CPU: 20ms
    CGroup: /system.slice/my_daemon.service
            └─748 /usr/local/bin/my_daemon

Oct 16 21:14:54 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 16 21:14:54 LAPTOP-JR1G744R my_daemon[748]: My daemon started
Oct 16 21:14:54 LAPTOP-JR1G744R my_daemon[748]: Daemon is running...
Oct 16 21:14:54 LAPTOP-JR1G744R systemd[1]: Started my_daemon.service - My Custom Daemon.
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo journalctl -u my_daemon
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:50 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 14 22:46:51 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 14 22:46:51 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 22:46:51 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:51 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:51 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
```



```

Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 4.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 22:46:58 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
-- Boot 4a2918601e8949f88e090e3f3848d770 --
Oct 15 17:21:58 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 15 17:21:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 15 17:21:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 15 17:21:58 LAPTOP-JR1G744R systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 15 17:21:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 15 17:21:58 LAPTOP-JR1G744R systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 15 17:21:58 LAPTOP-JR1G744R systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo systemctl stop my_daemon
nadiva_nathania@LAPTOP-JR1G744R:~$

```

Custom system service ini bertujuan pada sistem operasi seperti Linux, melalui system untuk mengelola dan menjalankan aplikasi atau skrip pengguna sebagai layanan latar belakang yang otomatis dan terintegrasi dengan sistem.

5. Linkers and Loaders

- Proses Linking

```

nadiva_nathania@LAPTOP-JR1G744R:~$ nano math_ops.h
nadiva_nathania@LAPTOP-JR1G744R:~$ nano math_ops.c
nadiva_nathania@LAPTOP-JR1G744R:~$ nano main.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -E main.c -o main.i
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -E math_ops.c -o math_ops.i
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -S main.c -o main.s
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -S math_ops.c -o math_ops.s
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -c main.c -o main.o
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -c math_ops.c -o math_ops.o
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc main.o math_ops.o -o program
nadiva_nathania@LAPTOP-JR1G744R:~$ nm main.o
                 U add
0000000000000000 T main
                 U multiply
                 U printf
nadiva_nathania@LAPTOP-JR1G744R:~$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
nadiva_nathania@LAPTOP-JR1G744R:~$ nm program
00000000000003dc8 d _DYNAMIC
00000000000003fb8 d _GLOBAL_OFFSET_TABLE_
00000000000002000 R _IO_stdin_used
                 w _ITM_deregisterTMCloneTable
                 w _ITM_registerTMCloneTable
00000000000002148 r __FRAME_END__
0000000000000201c r __GNU_EH_FRAME_HDR
00000000000004010 D __TMC_END__
0000000000000038c r __abi_tag
00000000000004010 B __bss_start
                 w __cxa_finalize@GLIBC_2.2.5
00000000000004000 D __data_start
00000000000001100 t __do_global_dtors_aux
00000000000003dc0 d __do_global_dtors_aux_fini_array_entry
00000000000004008 D __dso_handle
00000000000003db8 d __frame_dummy_init_array_entry
                 w __gmon_start__
                 U __libc_start_main@GLIBC_2.34

```

```

0000000000004010 D _edata
0000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
                                U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones
nadiva_nathania@LAPTOP-JR1G744R:~$ ldd program
        linux-vdso.so.1 (0x00007fff429dc000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x0000766f8ac00000)
        /lib64/ld-linux-x86-64.so.2 (0x0000766f8af11000)
nadiva_nathania@LAPTOP-JR1G744R:~$ |

```

Linking adalah tahap terakhir dalam proses kompilasi kode program. Tujuannya menyelesaikan semua referensi eksternal yang ada dalam kode objek sehingga program dapat berjalan dengan benar.

- **Static vs Dynamic Linking**

```

nadiva_nathania@LAPTOP-JR1G744R:~$ gcc main.c math_ops.c -o program_static -static
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc main.c math_ops.c -o program_dynamic
nadiva_nathania@LAPTOP-JR1G744R:~$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 16K Oct 15 19:10 program_dynamic
-rwxr-xr-x 1 nadiva_nathania nadiva_nathania 767K Oct 15 19:10 program_static
nadiva_nathania@LAPTOP-JR1G744R:~$ ldd program_static
        not a dynamic executable
nadiva_nathania@LAPTOP-JR1G744R:~$ ldd program_dynamic
        linux-vdso.so.1 (0x00007ffc78354000)
        libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007af014000000)
        /lib64/ld-linux-x86-64.so.2 (0x00007af01435e000)
nadiva_nathania@LAPTOP-JR1G744R:~$ |

```

Static linking adalah proses di mana linker menyalin semua kode dari library yang dibutuhkan biasanya berekstensi secara langsung ke dalam file program yang dapat dieksekusi. Sedangkan dynamic linking adalah proses di mana linker tidak menyalin kode library ke dalam executable. Sebaliknya, ia hanya menyimpan referensi atau catatan tentang library mana yang dibutuhkan, pemuatan dan penghubungan kode library yang sebenarnya baru terjadi saat program dimuat atau dijalankan.

- **Shared Library**

```

nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -c -fPIC math_ops.c -o math_ops.o
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -shared -o libmathops.so math_ops.o
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc main.c -L. -lmathops -o program_shared
nadiva_nathania@LAPTOP-JR1G744R:~$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15

```

```
nadiva_nathania@LAPTOP-JR1G744R:~$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
Tag                Type                Name/Value
0x0000000000000001 (NEEDED)           Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)           Shared library: [libc.so.6]
0x000000000000000c (INIT)             0x1000
0x000000000000000d (FINI)             0x11e4
0x0000000000000019 (INIT_ARRAY)          0x3d98
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)         0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x00000000000000f5 (GNU_HASH)           0x3b0
0x0000000000000005 (STRTAB)             0x4b0
0x0000000000000006 (SYMTAB)             0x3d8
0x000000000000000a (STRSZ)              170 (bytes)
0x000000000000000b (SYMENT)             24 (bytes)
0x0000000000000015 (DEBUG)              0x0
0x0000000000000003 (PLTGOT)             0x3fa8
0x0000000000000002 (PLTRELSZ)           72 (bytes)
0x0000000000000014 (PLTREL)             RELA
0x0000000000000017 (JMPREL)             0x660
0x0000000000000007 (RELA)              0x5a0
0x0000000000000008 (RELASZ)            192 (bytes)
0x0000000000000009 (RELAENT)           24 (bytes)
0x000000000000001e (FLAGS)             BIND_NOW
0x00000000000000fb (FLAGS_1)           Flags: NOW PIE
0x00000000000000fe (VERNEED)           0x570
0x00000000000000ff (VERNEEDNUM)        1
0x00000000000000f0 (VERSYM)            0x55a
0x00000000000000f9 (RELACOUNT)         3
0x0000000000000000 (NULL)              0x0
nadiva_nathania@LAPTOP-JR1G744R:~$ |
```

Shared library adalah sebuah file yang berisi kode dan data yang dapat dimuat di load ke memori dan digunakan bersama oleh beberapa program yang berjalan secara bersamaan, tujuannya agar penggunaan shared library meningkatkan efisiensi sistem dan pengembangan perangkat lunak.

6. Why Applications are Operating System Specific

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano os_specific.c
nadiva_nathania@LAPTOP-JR1G744R:~$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b47095af229368edc342f23076fc2, for GNU/Linux 3.2.0, not stripped
nadiva_nathania@LAPTOP-JR1G744R:~$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:             ELF64
  Data:              2's complement, little endian
  Version:           1 (current)
  OS/ABI:            UNIX - System V
  ABI Version:       0
  Type:              DYN (Position-Independent Executable file)
  Machine:           Advanced Micro Devices X86-64
  Version:           0x1
  Entry point address: 0x1060
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14072 (bytes into file)
  Flags:             0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
```

```
nadiva_nathania@LAPTOP-JR1G744R:~$ readelf -S program
There are 31 section headers, starting at offset 0x36f8:
```

Section Headers:

[Nr]	Name	Type	EntSize	Address	Flags	Link	Info	Offset	Align
Size									
[0]	0000000000000000	NULL	0000000000000000	0000000000000000	A	0	0	0	0
[1]	.interp	PROGBITS	0000000000000000	0000000000000318	A	0	0	1	0
[2]	.note.gnu.pr[...]	NOTE	0000000000000000	0000000000000338	A	0	0	8	0
[3]	.note.gnu.bu[...]	NOTE	0000000000000000	0000000000000368	A	0	0	4	0
[4]	.note.ABI-tag	NOTE	0000000000000000	000000000000038c	A	0	0	1	0
[5]	.gnu.hash	GNU_HASH	0000000000000000	00000000000003b0	A	6	0	8	0
[6]	.dynsym	DYNSYM	0000000000000000	00000000000003d8	A	7	1	8	0
[7]	.dynstr	STRTAB	0000000000000000	0000000000000480	A	0	0	1	0
[8]	.gnu.version	VERSYM	0000000000000000	0000000000000510	A	6	0	2	0
[9]	.gnu.version_r	VERNEED	0000000000000000	0000000000000520	A	7	1	8	0
[10]	.rela.dyn	RELA	0000000000000000	0000000000000550	A	6	0	8	0
[11]	.rela.plt	RELA	0000000000000000	0000000000000610	AI	6	24	8	0
[12]	.init	PROGBITS	0000000000000000	0000000000001000	AX	0	0	4	0
[13]	.plt	PROGBITS	0000000000000000	0000000000001020	AX	0	0	16	0
[14]	.plt.got	PROGBITS	0000000000000000	0000000000001040	AX	0	0	16	0
[15]	.plt.sec	PROGBITS	0000000000000000	0000000000001050	AX	0	0	16	0

[15]	.plt.sec	PROGBITS	0000000000001050	00001050	AX	0	0	16	0
[16]	.text	PROGBITS	0000000000000010	00001060	AX	0	0	16	0
[17]	.fini	PROGBITS	000000000000000d	000011d4	AX	0	0	16	0
[18]	.rodata	PROGBITS	0000000000000000	00002000	A	0	0	4	0
[19]	.eh_frame_hdr	PROGBITS	000000000000001c	0000201c	A	0	0	4	0
[20]	.eh_frame	PROGBITS	0000000000000044	00002060	A	0	0	8	0
[21]	.init_array	INIT_ARRAY	00000000000000ec	00002db8	WA	0	0	8	0
[22]	.fini_array	FINI_ARRAY	0000000000000008	00002dc0	WA	0	0	8	0
[23]	.dynamic	DYNAMIC	0000000000000008	00003dc8	WA	0	0	8	0
[24]	.got	PROGBITS	00000000000001f0	00003fb8	WA	7	0	8	0
[25]	.data	PROGBITS	0000000000000048	00003000	WA	0	0	8	0
[26]	.bss	NOBITS	0000000000000010	00003010	WA	0	0	8	0
[27]	.comment	PROGBITS	0000000000000008	00003010	WA	0	0	1	0
[28]	.symtab	SYMTAB	000000000000002b	00003040	MS	0	0	1	0
[29]	.strtab	STRTAB	00000000000003a8	000030e8	MS	29	19	8	0
[30]	.shstrtab	STRTAB	00000000000001f4	000035dc	0	0	0	1	0
[31]			000000000000011a	000035dc	0	0	0	1	0

Key to Flags:

W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
L (link order), O (extra OS processing required), G (group), T (TLS),
C (compressed), x (unknown), o (OS specific), E (exclude),
D (mbind), l (large), p (processor specific)

```
nadiva_nathania@LAPTOP-JR1G744R:~$ readelf -l program
```

Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1060
There are 13 program headers, starting at offset 64

Program Headers:

Type	Offset	VirtAddr	PhysAddr
FileSiz	MemSiz	Flags	Align
PHDR	0x0000000000000040	0x0000000000000040	0x0000000000000040
INTERP	0x00000000000002d8	0x00000000000002d8	R 0x8
LOAD	0x0000000000000318	0x0000000000000318	0x0000000000000318
LOAD	0x000000000000001c	0x000000000000001c	R 0x1
[Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]			
LOAD	0x0000000000000000	0x0000000000000000	0x0000000000000000
LOAD	0x0000000000000628	0x0000000000000628	R 0x1000
LOAD	0x0000000000001000	0x0000000000001000	0x0000000000001000
LOAD	0x0000000000001e1	0x0000000000001e1	R E 0x1000
LOAD	0x0000000000002000	0x0000000000002000	0x0000000000002000
LOAD	0x00000000000014c	0x00000000000014c	R 0x1000
LOAD	0x0000000000002db8	0x0000000000003db8	0x0000000000003db8
DYNAMIC	0x000000000000258	0x000000000000260	RW 0x1000
NOTE	0x0000000000002dc8	0x0000000000003dc8	0x0000000000003dc8
NOTE	0x0000000000001f0	0x0000000000001f0	RW 0x8
NOTE	0x0000000000000338	0x0000000000000338	0x0000000000000338
NOTE	0x0000000000000030	0x0000000000000030	R 0x8
NOTE	0x0000000000000368	0x0000000000000368	0x0000000000000368
GNU_PROPERTY	0x0000000000000044	0x0000000000000044	R 0x4
GNU_PROPERTY	0x0000000000000338	0x0000000000000338	0x0000000000000338
GNU_EH_FRAME	0x0000000000000030	0x0000000000000030	R 0x8
GNU_EH_FRAME	0x000000000000201c	0x000000000000201c	0x000000000000201c
GNU_STACK	0x0000000000000044	0x0000000000000044	R 0x4
GNU_STACK	0x0000000000000000	0x0000000000000000	0x0000000000000000
GNU_RELRO	0x0000000000000000	0x0000000000000000	RW 0x10
GNU_RELRO	0x0000000000002db8	0x0000000000003db8	0x0000000000003db8
GNU_RELRO	0x0000000000000248	0x000000000000248	R 0x1

Section to Segment mapping:

Segment Sections...

00	
01	.interp
02	.interp .note.gnu.property .note.gnu.build-id .note.ABI-tag .gnu.hash .dynsym .dynstr .gnu.version .gnu.version_r .rela.dyn
.rela.plt	
03	.init .plt .plt.got .plt.sec .text .fini
04	.rodata .eh_frame_hdr .eh_frame
05	.init_array .fini_array .dynamic .got .data .bss
06	.dynamic
07	.note.gnu.property
08	.note.gnu.build-id .note.ABI-tag
09	.note.gnu.property
10	.eh_frame_hdr
11	
12	.init_array .fini_array .dynamic .got

```
nadiva_nathania@LAPTOP-JR1G744R:~$ objdump -d program | head -50

program:      file format elf64-x86-64

Disassembly of section .init:

0000000000001000 <.init>:
1000:      f3 0f 1e fa          endbr64
1004:      48 83 ec 08          sub     $0x8,%rsp
1008:      48 b0 05 d9 2f 00 00  mov     0x2fd9(%rip),%rax      # 3fe8 <__gmon_start__@Base>
100f:      48 85 c0             test    %rax,%rax
1012:      74 02             je      1016 <_init+0x16>
1014:      ff d0             call    *%rax
1016:      48 83 c4 08          add     $0x8,%rsp
101a:      c3             ret

Disassembly of section .plt:

0000000000001020 <.plt>:
1020:      ff 35 9a 2f 00 00    push    0x2f9a(%rip)          # 3fc0 <_GLOBAL_OFFSET_TABLE_+0x8>
1026:      ff 25 9c 2f 00 00    jmp     *0x2f9c(%rip)          # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x10>
102c:      0f 1f 40 00          nopl     0x0(%rax)
1030:      f3 0f 1e fa          endbr64
1034:      68 00 00 00 00 00    push    $0x0
1039:      e9 e2 ff ff          jmp     1020 <_init+0x20>
103e:      66 90             xchg     %ax,%ax

Disassembly of section .plt.got:

0000000000001040 <_cxa_finalize@plt>:
1040:      f3 0f 1e fa          endbr64
1044:      ff 25 ae 2f 00 00    jmp     *0x2fae(%rip)          # 3ff8 <_cxa_finalize@GLIBC_2.2.5>
104a:      66 0f 1f 44 00 00    nopw     0x0(%rax,%rax,1)

Disassembly of section .plt.sec:
```

- **System Call Differences**

Tujuannya adalah untuk memungkinkan setiap OS mengoptimalkan kinerja, mengelola hardware yang unik, dan menerapkan model keamanan serta fitur-fitur eksklusifnya secara maksimal.

- **Binary Format Analysis**

Binary format analysis adalah proses pemeriksaan, interpretasi, dan dekonstruksi struktur internal dari sebuah file biner, tujuannya adalah untuk memahami bagaimana data disusun, diwakili, dan diproses oleh program atau sistem komputer.

7. OS Design and Implementation

- **Kernel Module Sederhana**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano hello_module.c
nadiva_nathania@LAPTOP-JR1G744R:~$ nano Makefile
nadiva_nathania@LAPTOP-JR1G744R:~$ make
sudo insmod hello_module.ko
lsmod | grep hello
dmesg | tail
sudo rmmod hello_module
dmesg | tail
make -C /lib/modules/6.8.7.2-microsoft-standard-WSL2/build M=/home/nadiva_nathania/modules
make[1]: *** /lib/modules/6.8.7.2-microsoft-standard-WSL2/build: No such file or directory. Stop.
make: *** [Makefile:4: all] Error 2
[sudo] password for nadiva_nathania:
insmod: ERROR: could not load module hello_module.ko: No such file or directory
[ 2.529386] kvm_intel: Using Hyper-V Enlightened VMCS
[ 2.701410] intel_rapl_msr: PL4 support detected.
[ 4.274317] systemd-journald[45]: /var/log/journal/8228cb8774e544abb6d2cf183e54a713/user-1000.journal: Journal file uses a different
sequence number ID, rotating.
[ 6.058553] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 48.937292] hv_balloon: Max. dynamic memory size: 3926 MB
[ 615.844722] mini_init (223): drop_caches: 1
[ 985.919777] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 1367.968070] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 1637.924465] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 7336.898065] mini_init (223): drop_caches: 1
rmmod: ERROR: Module hello_module is not currently loaded
[ 2.529386] kvm_intel: Using Hyper-V Enlightened VMCS
[ 2.701410] intel_rapl_msr: PL4 support detected.
[ 4.274317] systemd-journald[45]: /var/log/journal/8228cb8774e544abb6d2cf183e54a713/user-1000.journal: Journal file uses a different
sequence number ID, rotating.
[ 6.058553] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 48.937292] hv_balloon: Max. dynamic memory size: 3926 MB
[ 615.844722] mini_init (223): drop_caches: 1
[ 985.919777] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 1367.968070] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 1637.924465] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 7336.898065] mini_init (223): drop_caches: 1
```


Kernel module sederhana adalah file objek yang berisi kode yang dapat di load dan unload dari Kernel sistem operasi yang sedang berjalan, tujuannya agar penggunaan dan pembuatan kernel modul adalah untuk mencapai efisiensi sumber daya dan ekstensibilitas sistem operasi.

- **Monitoring Kernel**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ uname -r
6.6.87.2-microsoft-standard-WSL2
nadiva_nathania@LAPTOP-JR1G744R:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNA
MIC Thu Jun 5 18:58:46 UTC 2025
nadiva_nathania@LAPTOP-JR1G744R:~$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall32 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autosubject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
```

```
nadiva_nathania@LAPTOP-JR1G744R:~$ dmesg | tail -30
[ 1.067320] scsi 0:0:0:3: Direct-Access Msft Virtual Disk 1.0 PQ: 0 ANSI: 5
[ 1.074972] sd 0:0:0:3: Attached scsi generic sg2 type 0
[ 1.075532] sd 0:0:0:3: [sdd] 2107483648 512-byte logical blocks: (1.10 TB/1.00 TiB)
[ 1.075999] sd 0:0:0:3: [sdd] 4096-byte physical blocks
[ 1.076421] sd 0:0:0:3: [sdd] Write Protect is off
[ 1.077257] sd 0:0:0:3: [sdd] Mode Sense: 0f 00 00 00
[ 1.077823] sd 0:0:0:3: [sdd] Write cache: enabled, read cache: enabled, doesn't support DPO or FUA
[ 1.080234] sd 0:0:0:3: [sdd] Attached SCSI disk
[ 1.101280] EXT4-fs (sdd): mounted filesystem 0alad939-a16c-47c4-99ab-af61af718107 r/w with ordered data mode. Quota mode: none.
[ 1.621493] pulseaudio[275]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 1.760692] misc dxg: dxgk: dxgkio.is.feature.enabled: Ioctl failed: -22
[ 1.764725] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.765376] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.765880] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.766576] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 1.996792] systemd-journald[45]: Collecting audit messages is disabled.
[ 2.092911] systemd-journald[45]: Received client request to flush runtime journal.
[ 2.096543] systemd-journald[45]: File /var/log/journal/8228cb8774e544abb6d2cf183e54a713/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 2.268487] ACPI: AC: AC Adapter [AC1] (off-line)
[ 2.272647] ACPI: battery: Slot [BAT1] (battery present)
[ 2.529386] kvm_intel: Using Hyper-V Enlightened VMCS
[ 2.701414] intel_rapl_msr: PL4 support detected.
[ 4.274317] systemd-journald[45]: /var/log/journal/8228cb8774e544abb6d2cf183e54a713/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 6.058553] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 48.937292] hv.balloon: Max. dynamic memory size: 3926 MB
[ 615.844722] mini_init (223): drop_caches: 1
[ 985.919777] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 1367.968070] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 1637.924465] WSL (239) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 7336.898065] mini_init (223): drop_caches: 1
```

```
nadiva_nathania@LAPTOP-JR1G744R:~$ lsmod
Module      Size  Used by
tls          98304  0
intel_rapl_msr 16384  0
intel_rapl_common 32768  1 intel_rapl_msr
kvm_intel    356352  0
kvm          970752  1 kvm_intel
irqbypass   12288  1 kvm
crc32c_intel 16384  0
battery      20480  0
ac           16384  0
sch_fq_codel 16384  1
configfs     53248  1
autofs4      45056  0
br_netfilter 28672  0
bridge       282624  1 br_netfilter
stp          12288  1 bridge
llc          12288  2 bridge,stp
ip_tables    28672  0
tun          53248  0
nadiva_nathania@LAPTOP-JR1G744R:~$ modinfo ext4
name: ext4
filename: (builtin)
softdep: pre: crc32c
license: GPL
file: fs/ext4/ext4
description: Fourth Extended Filesystem
author: Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias: fs-ext4
alias: ext3
alias: fs-ext3
```

Monitoring kernel adalah proses pengumpulan, analisis, dan visualisasi data tentang aktivitas, kinerja, dan status dari kernel sistem operasi, tujuannya adalah untuk memastikan stabilitas, keamanan, dan efisiensi sistem secara keseluruhan.

8. Operating System Structure

- **Eksplorasi Struktur Layering**

Layer 1: Hardware

```

nvidia_nathania@LAPTOP--JR1G7UAR:~$ lspci
5582:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio 1.0 console (rev 01)
d71a:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
c35:00:00.0 System peripheral: Red Hat, Inc. Virtio file system (rev 01)
nvidia_nathania@LAPTOP--JR1G7UAR:~$ lsusb
nvidia_nathania@LAPTOP--JR1G7UAR:~$ cat /proc/cpuinfo
processor           : 0
vendor_id          : GenuineIntel
cpu family         : 6
model              : 186
model name         : 13th Gen Intel(R) Core(TM) i5-13420H
stepping           : 2
microcode          : 0xffffffffff
cpu MHz            : 2611.199
cache size         : 12288 KB
physical id        : 0
siblings           : 12
core id            : 0
cpu cores          : 6
apicid             : 0
initial apicid     : 0
fpu                : yes
fpu_exception      : yes
cpuid level        : 28
wp                 : yes
flags              : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx
                        pdgclb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma c
pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs
                        brb sha3_ni xsaveopt xsavec xgetbv1 xsave avx_vnni vmmi waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fwm md_clear seri
vex_flags_l1d arch_capabilities
vnx flags          : vnni invnvp ept_x_only ept_aid ept_lgb tsc_offset vtptr ept_vmx unrestricted_guest ept_mode_based_exec tsc_scaling
user_wait_pause    :
bugs               : spectre_v1 spectre_v2 spec_store_bypass swapgs retbleed eibrs_pbrsb rfdbs bhi
bogomips           : 5222.39
clflush_size       : 64
cache_alignment    : 64

```

Layer 2: Kernel

```
nadiva_nathania@LAPTOP-JR1G744R:~$ uname -a
Linux LAPTOP-JR1G744R 6.8.67.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
nadiva_nathania@LAPTOP-JR1G744R:~$ cat /proc/sys/kernel/ostype
linux
```

Layer 3: System Calls

[illegible]

Layer 4: System Programs

```
nadiva_nathania@LAPTOP-JR1G744R:~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.0  0.3 21836 12192 ?        Ss   09:08   0:01 /sbin/init
root             2  0.0  0.0   3072  1664 ?        Sl   09:08   0:00 /init
root             8  0.0  0.0   3072  1792 ?        Sl   09:08   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10
root          45  0.0  0.4 66812 18960 ?        S<s  09:08   0:00 /usr/lib/systemd/systemd-journald
root          94  0.0  0.1 25264  6272 ?        Ss   09:08   0:00 /usr/lib/systemd/systemd-udev
systemd+     110  0.0  0.3 21456 12032 ?        Ss   09:08   0:00 /usr/lib/systemd/systemd-resolved
systemd+     111  0.0  0.1 91024  7424 ?        Ssl  09:08   0:00 /usr/lib/systemd/systemd-timesyncd
root         162  0.0  0.0  4236  2560 ?        Ss   09:08   0:00 /usr/sbin/cron -f -P
message+    163  0.0  0.1  9624  4736 ?        Ss   09:08   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --sys
root        180  0.0  0.2 17964  8064 ?        Ss   09:08   0:00 /usr/lib/systemd/systemd-logind
root        184  0.0  0.3 1756096 12160 ?       Ssl  09:08   0:00 /usr/libexec/wsl-pro-service -vv
root        196  0.0  0.0   3160  1920 hvc0     Ss+  09:08   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600
syslog      206  0.0  0.1 222508  5504 ?        Ssl  09:08   0:00 /usr/sbin/rsyslogd -n -iNONE
root        221  0.0  0.0   3116  1792 ttyL     Ss+  09:08   0:00 /sbin/agetty -o -p -- \u --noclear - linux
root        228  0.0  0.5 107012 22516 ?       Ssl  09:08   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade
root        357  0.0  0.1  6664  4352 pts/1    Ss   09:08   0:00 /bin/login -f
nadiva_+    403  0.0  0.2 20316 11264 ?        Ss   09:08   0:00 /usr/lib/systemd/systemd --user
nadiva_+    404  0.0  0.0 21152  3516 ?        S   09:08   0:00 (sd-pam)
nadiva_+    426  0.0  0.1  6072  4992 pts/1    S+   09:08   0:00 -bash
root        506  0.0  0.0   3076  1028 ?        Ss   09:08   0:00 /init
root        507  0.0  0.0   3092  1032 ?        S   09:08   0:00 /init
nadiva_+    510  0.0  0.1  6200  5248 pts/0    Ss   09:08   0:00 -bash
nadiva_+    662 400  0.1  8280  4096 pts/0    Rr   09:59   0:00 ps aux
```

```
nadiva_nathania@LAPTOP-JR1G744R:~$ top
top - 09:59:45 up 51 min, 1 user, load average: 0.06, 0.02, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 3747.2 total, 3211.9 free, 432.7 used, 177.1 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used, 3314.5 avail Mem

   PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21836  12192  9248 S   0.3   0.3   0:01.02 systemd
   111 systemd+  20   0   91024  7424   6656 S   0.3   0.2   0:00.21 systemd-timesyn
   507 root       20   0   3092   1032   896 S   0.3   0.0   0:00.04 Relay(510)
   663 nadiva_+   20   0   9284   5504   3328 R   0.3   0.1   0:00.02 top
     2 root       20   0   3072   1664   1664 S   0.0   0.0   0:00.01 init-systemd(Ub
     8 root       20   0   3072   1792   1792 S   0.0   0.0   0:00.00 init
    45 root      19  -1  66812  18960  18192 S   0.0   0.5   0:00.51 systemd-journal
    94 root       20   0   25264  6272   4864 S   0.0   0.2   0:00.34 systemd-udev
   110 systemd+  20   0   21456  12032  10112 S   0.0   0.3   0:00.16 systemd-resolve
   162 root       20   0   4236   2560   2432 S   0.0   0.1   0:00.02 cron
   163 message+   20   0   9624   4736   4352 S   0.0   0.1   0:00.21 dbus-daemon
   180 root       20   0   17964  8064   7296 S   0.0   0.2   0:00.12 systemd-logind
   184 root       20   0 1756096 12160   10240 S   0.0   0.3   0:00.29 wsl-pro-service
   196 root       20   0   3160   1920   1792 S   0.0   0.1   0:00.01 agetty
   206 syslog    20   0 222508  5504   4480 S   0.0   0.1   0:00.13 rsyslogd
   221 root       20   0   3116   1664   1664 S   0.0   0.0   0:00.00 agetty
   228 root       20   0 107012 22516  13312 S   0.0   0.6   0:00.13 unattended-upgr
   357 root       20   0   6664   4352   3712 S   0.0   0.1   0:00.00 login
   403 nadiva_+   20   0 20316  11264   9216 S   0.0   0.3   0:00.12 systemd
   404 nadiva_+   20   0 21152  3516   1792 S   0.0   0.1   0:00.00 (sd-pam)
   426 nadiva_+   20   0  6072  4992   3456 S   0.0   0.1   0:00.01 bash
   506 root       20   0   3076   1028   896 S   0.0   0.0   0:00.00 SessionLeader
   510 nadiva_+   20   0  6200  5248   3584 S   0.0   0.1   0:00.07 bash
```

Layer 5: User Applications

```
nadiva_nathania@LAPTOP-JR1G744R:~$ which bash
/usr/bin/bash
nadiva_nathania@LAPTOP-JR1G744R:~$ which gcc
/usr/bin/gcc
nadiva_nathania@LAPTOP-JR1G744R:~$ |
```

Eksplorasi struktur layering merujuk pada proses analisis dan pemahaman bagaimana sebuah sistem kompleks diorganisasikan menjadi lapisan-lapisan yang terpisah dan saling berinteraksi, tujuan dari eksplorasi ini adalah untuk memahami prinsip-prinsip desain, fungsionalitas, dan ketergantungan antara komponen sistem tersebut.

- **Microkernel vs Monolithic**

Lihat ukuran kernel (Monolithic)

```
nadiva_nathania@LAPTOP-JR1G744R:~$ uname -r
6.6.87.2-microsoft-standard-WSL2
```

Lihat built-in vs module

```
nadiva_nathania@LAPTOP-JR1G744R:~$ uname -r | grep "=y" | wc -l
0
nadiva_nathania@LAPTOP-JR1G744R:~$ uname -r | grep "=m" | wc -l
0
```

Lihat loaded modules

```
nadiva_nathania@LAPTOP-JR1G744R:~$ lsmod | wc -l
19
nadiva_nathania@LAPTOP-JR1G744R:~$ |
```

Monolithic kernel adalah arsitektur di mana semua layanan inti OS berada dalam satu blok kode tunggal, berjalan di ruang kernel dengan hak akses tertinggi. Sedangkan microkernel adalah arsitektur di mana kernel diminimalkan hanya untuk layanan inti mutlak, sebagian besar layanan OS tradisional dipindahkan ke ruang pengguna dan berjalan sebagai program terpisah yang disebut server

- **Program Virtual File System.**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ nano procfs_read.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -o procfs_read procfs_read.c
nadiva_nathania@LAPTOP-JR1G744R:~$ ./procfs_read
=== CPU Information ===
processor      : 0
vendor_id     : GenuineIntel
cpu family    : 6
model         : 186
model name    : 13th Gen Intel(R) Core(TM) i5-13420H
stepping      : 2
microcode     : 0xffffffff
cpu MHz       : 2611.199
cache size    : 12288 KB
physical id   : 0

=== Memory Information ===
MemTotal:     3837140 kB
MemFree:      3162720 kB
MemAvailable: 3388712 kB
Buffers:      10748 kB
Cached:       269948 kB

=== Process Information ===
Name:  procfs_read
Umask: 0022
State: R (running)
Tgid:  1194
Ngid:  0
Pid:   1194
PPid:  510
TracerPid: 0
Uid:   1000    1000    1000    1000
Gid:   1000    1000    1000    1000
nadiva_nathania@LAPTOP-JR1G744R:~$ |
```

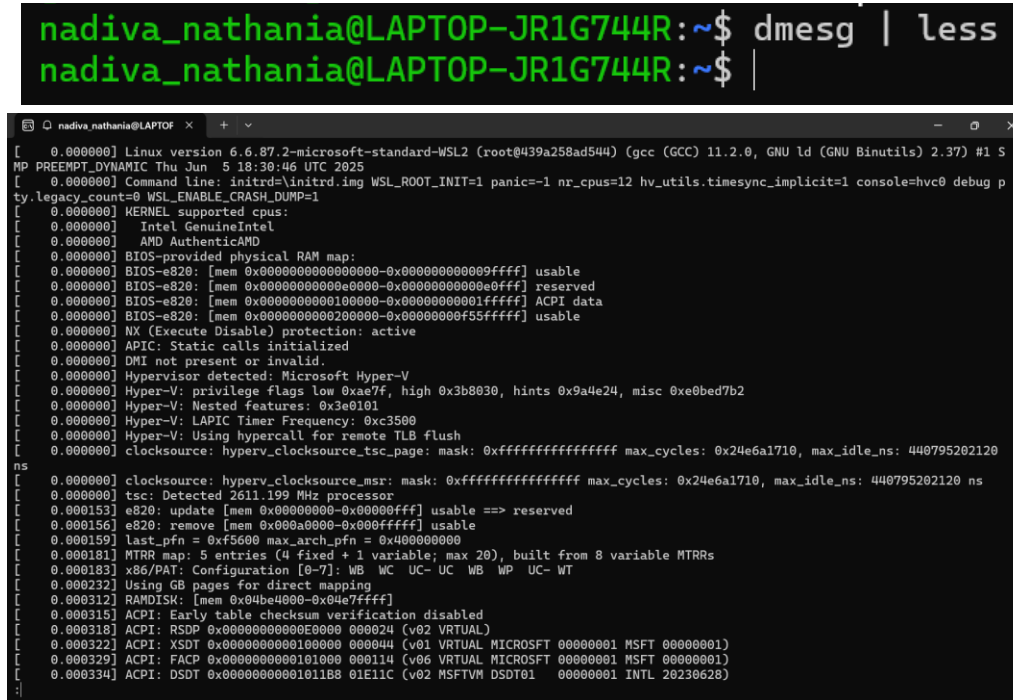
Program virtual file system adalah lapisan perangkat lunak dalam kernel sistem operasi yang menyediakan antarmuka terpadu dan dapat mengakses berbagai jenis sistem berkas,

tujuannya adalah untuk memungkinkan program aplikasi berinteraksi dengan berkas, terlepas dari di mana berkas itu disimpan atau format sistem berkasnya.

9. Building and Booting an Operating System

- **Analisis Boot Process**

Lihat boot messages



The image shows a terminal window with a dark background. The top part shows a prompt for 'nadiva_nathania@LAPTOP-JR1G744R' where the command 'dmesg | less' has been entered. Below this, a window titled 'nadiva_nathania@LAPTOP' displays the output of the command. The output is a list of boot messages from the Linux kernel, including information about the kernel version (6.6.87.2-microsoft-standard-WSL2), the command line, hardware details like CPU and memory, and various system initialization steps like BIOS, ACPI, and hypervisor detection.

```
nadiva_nathania@LAPTOP-JR1G744R:~$ dmesg | less
nadiva_nathania@LAPTOP-JR1G744R:~$

[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 S
MP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug p
ty_legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000]   Intel GenuineIntel
[ 0.000000]   AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000a0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x0000000000f5ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x9a4e24, misc 0xe0bed7b2
[ 0.000000] Hyper-V: Nested features: 0x3e0101
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120
ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 2611.199 MHz processor
[ 0.001533] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.001563] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.001593] last_pfn = 0xf5600 max_arch_pfn = 0x400000000
[ 0.001811] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[ 0.001831] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
[ 0.002321] Using GB pages for direct mapping
[ 0.003121] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[ 0.003151] ACPI: Early table checksum verification disabled
[ 0.003181] ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
[ 0.003221] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.003291] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.003334] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
```

Lihat systemd boot analysis

```
nadiva_nathania@LAPTOP-JR1G744R:~$ systemd-analyze
Startup finished in 2.045s (userspace)
graphical.target reached after 1.994s in userspace.
nadiva_nathania@LAPTOP-JR1G744R:~$ systemd-analyze blame
1.322s landscape-client.service
373ms snapd.seeded.service
288ms dev-sdd.device
281ms snapd.service
264ms wsl-pro.service
175ms user@1000.service
144ms systemd-resolved.service
125ms systemd-tmpfiles-clean.service
116ms rsyslog.service
97ms systemd-udev-trigger.service
89ms systemd-timesyncd.service
82ms systemd-logind.service
79ms systemd-journal-flush.service
60ms keyboard-setup.service
60ms systemd-journald.service
55ms systemd-udev.service
42ms dbus.service
38ms e2scrub_reap.service
37ms sys-kernel-tracing.mount
37ms systemd-tmpfiles-setup.service
36ms dev-hugepages.mount
36ms sys-kernel-debug.mount
35ms dev-mqueue.mount
30ms systemd-update-utmp.service
29ms kmod-static-nodes.service
28ms modprobe@configfs.service
28ms modprobe@dm_mod.service
27ms modprobe@drm.service
27ms systemd-tmpfiles-setup-dev-early.service
27ms modprobe@efi_pstore.service
26ms modprobe@fuse.service
26ms systemd-sysctl.service
23ms modprobe@loop.service
```

```

37ms sys-kernel-tracing.mount
37ms systemd-tmpfiles-setup.service
36ms dev-hugepages.mount
36ms sys-kernel-debug.mount
35ms dev-mqueue.mount
30ms systemd-update-utmp.service
29ms kmod-static-nodes.service
28ms modprobe@configfs.service
28ms modprobe@dm_mod.service
27ms modprobe@drm.service
27ms systemd-tmpfiles-setup-dev-early.service
27ms modprobe@efi_pstore.service
26ms modprobe@fuse.service
26ms systemd-sysctl.service
23ms modprobe@loop.service
23ms user-runtime-dir@1000.service
21ms systemd-modules-load.service
20ms systemd-remount-fs.service
nadiva_nathania@LAPTOP-JR1G744R:~$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @1.994s
├─multi-user.target @1.992s
│   └─snapd.seeded.service @684ms +373ms
│       └─basic.target @650ms
│           └─sockets.target @650ms
│               └─snapd.socket @640ms +8ms
│                   └─sysinit.target @630ms
│                       └─systemd-resolved.service @485ms +144ms
│                           └─systemd-tmpfiles-setup.service @410ms +37ms
│                               └─systemd-journal-flush.service @327ms +79ms
│                                   └─systemd-journald.service @250ms +60ms
│                                       └─systemd-journald.socket @239ms
│                                           └─.mount @220ms
│                                               └─.slice @220ms

```

Lihat GRUB configuration

```

nadiva_nathania@LAPTOP-JR1G744R:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
nadiva_nathania@LAPTOP-JR1G744R:~$ |

```

Lihat initramfs

```

nadiva_nathania@LAPTOP-JR1G744R:~$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
/usr/bin/unmkinitramfs: 64: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 57: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
/usr/bin/unmkinitramfs: 38: cannot open /boot/initrd.img-6.6.87.2-microsoft-standard-WSL2: No such file
cpio: premature end of archive
nadiva_nathania@LAPTOP-JR1G744R:~$ |

```

Analisis boot process adalah proses pemeriksaan, pemahaman, dan evaluasi langkah-langkah berurutan yang dilakukan oleh sistem komputer sejak dinyalakan hingga sistem operasi (OS) siap digunakan. tujuannya adalah untuk memecahkan masalah, mengoptimalkan waktu boot, dan mengidentifikasi keamanan.

- **Kernel Parameters**

Lihat kernel parameters saat boot

```

nadiva_nathania@LAPTOP-JR1G744R:~$ cat /proc/cmdline
initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_
CRASH_DUMP=1
nadiva_nathania@LAPTOP-JR1G744R:~$ |

```

Edit GRUB untuk menambah parameter (temporary)

Saat boot, tekan 'e' di GRUB menu, tambahkan parameter

```
# sudo update-grub
```

- **Simple Bootloader**

10. Operating System Debugging

- ## # Trace system calls

Trace dengan detail

ing waktu system calls

```
# Trace dan simpan ke file
```

[illegible]

Strace digunakan untuk memonitor interaksi antara suatu proses dan kernel sistem operasi, secara spesifik, strace melacak dan mencatat semua system call yang dilakukan oleh suatu program, tujuan utama menggunakan strace adalah untuk debugging, analisis kinerja, dan pemahaman tingkat rendah tentang bagaimana suatu program berinteraksi dengan lingkungannya.

- **Using ltrace**

```
# Trace library calls
```

```

nvidia.nathan@xbl-APT0P-JR16744UR:~$ ltrace ls
strchr("ls", '/')          = nil
setlocale(LC_ALL, "")       = "C.UTF-8"
bindtextdomain("coreutils", "/usr/share/locale")
textdomain("coreutils")     = "coreutils"
__cxa_atexit(0x64bdcf3de7880, 0, 0x64bdcf3df0f88, 0)
getopt_long(1, 0x7ffe3e196c28, "abcdcfghiklmnopqrstuvw:ABCDCFGHI:... , 0x64bdcf3dfe340, -1) = -1
getenv("LS_BLOCK_SIZE")    = nil
getenv("BLOCK_SIZE")       = nil
getenv("BLOCK_SIZE")       = nil
getenv("POSIXLY_CORRECT")   = nil
getenv("BLOCK_SIZE")       = nil
isatty(1)                   = 1
ioctl(1, 21523, 0x7ffe3e196a40)
getenv("TABSIZE")          = nil
getenv("QUOTING_STYLE")     = nil
__errno_location()          = 0x775ef04256a0
__errno_location()          = 0x775ef04256a0
getenv("TZ")                = nil
reallocarray(0, 100, 208, 0x64bd21aa16c0)
strlen(".")                 = 1
__mempcy_chk(0x64bd21aa6850, 0x64bdcf3df6dec, 2, 2)
__errno_location()          = 0x64bd21aa16d0
opendir(".")                = 0x64bd21aa6850
readdir(0x64bd21aa6870)     = 0x64bd21aa6870
readdir(0x64bd21aa6870)     = 0x64bd21aa68a0
__errno_location()          = 0x64bd21aa68c0
__ctype_get_mb_cur_max()    = 6
strlen("test_dir")          = 8
strlen("test_dir")          = 8
__mempcy_chk(0x64bd21aa8b0, 0x64bd21aa68d3, 9, 9)
readdir(0x64bd21aa6870)     = 0x64bd21aa68e0
__errno_location()          = 0x64bd21aa68e0
__ctype_get_mb_cur_max()    = 6
strlen("procfs_read.c")     = 13
strlen("procfs_read.c")     = 13
__mempcy_chk(0x64bd21aa8d0, 0x64bd21aa68f3, 14, 14)

```

Dengan detail

```
nadiva_nathania@LAPTOP-JR1G744R:~$ ltrace -c ls
Makefile      hello_module.c  math_ops.c      my_daemon.c      praktikum_os      program_shared    test.txt
bootloader_concept.asm  info_syscalls.c  math_ops.h      myshell.c         praktikum_os      program_static    test_dir
buggy          info_syscalls.c  math_ops.i      myshell.c         process_syscalls  simple_shell      test_file.txt
buggy.c        libmathops.so    math_ops.o      myshell.project   process_syscalls.c  simple_shell.c    trace.txt
file.txt       main.c           math_ops.s      os_services       procfs_read       syscall_test.txt
file_syscalls  main.i           memleak.c       os_services.c     procfs_read.c     sysmon
file_syscalls.c  main.o           memleak.c       os_specific.c     program            sysmon.c
files.txt       main.s           my_daemon       out.txt           program_dynamic    sysnom.c

% time      seconds  usecs/call  calls  function
-----
30.94 0.057917 200 289 __errno_location
27.68 0.051814 221 234 strcoll
10.75 0.020129 96 289 strlen
7.40 0.014010 93 150 __overflow
6.79 0.012704 81 156 __ctype_get_mb_cur_max
3.22 0.006036 92 65 readdir
3.13 0.005864 5864 1 setlocale
2.88 0.005391 224 24 memcpy
2.53 0.004736 89 53 __memcpy_chk
2.35 0.004396 84 52 fwrite_unlocked
0.36 0.000665 166 4 __freading
0.32 0.000608 76 8 getenv
0.20 0.000378 94 4 reallocarray
0.19 0.000354 177 2 fflush
0.18 0.000335 167 2 __fpending
0.17 0.000319 159 2 fclose
0.15 0.000282 141 2 fileno
0.11 0.000204 204 1 ioctl
0.10 0.000178 178 1 strrchr
0.09 0.000176 176 1 opendir
0.06 0.000120 120 1 isatty
0.06 0.000119 119 1 closedir
0.06 0.000118 118 1 bindtextdomain
0.05 0.000102 102 1 setjmp
0.05 0.000097 97 1 __cxa_atexit
0.03 0.000062 62 1 getopt_long

% time      seconds  usecs/call  calls  function
-----
30.94 0.057917 200 289 __errno_location
27.68 0.051814 221 234 strcoll
10.75 0.020129 96 289 strlen
7.40 0.014010 93 150 __overflow
6.79 0.012704 81 156 __ctype_get_mb_cur_max
3.22 0.006036 92 65 readdir
3.13 0.005864 5864 1 setlocale
2.88 0.005391 224 24 memcpy
2.53 0.004736 89 53 __memcpy_chk
2.35 0.004396 84 52 fwrite_unlocked
0.36 0.000665 166 4 __freading
0.32 0.000608 76 8 getenv
0.20 0.000378 94 4 reallocarray
0.19 0.000354 177 2 fflush
0.18 0.000335 167 2 __fpending
0.17 0.000319 159 2 fclose
0.15 0.000282 141 2 fileno
0.11 0.000204 204 1 ioctl
0.10 0.000178 178 1 strrchr
0.09 0.000176 176 1 opendir
0.06 0.000120 120 1 isatty
0.06 0.000119 119 1 closedir
0.06 0.000118 118 1 bindtextdomain
0.05 0.000102 102 1 setjmp
0.05 0.000097 97 1 __cxa_atexit
0.03 0.000062 62 1 getopt_long
0.03 0.000058 58 1 textdomain
-----
100.00 0.187172 1267 total
```

ltrace adalah sistem operasi untuk melacak panggilan dan eksekusi fungsi di dalam sebuah program yang berada di user space. Secara khusus, ltrace ini mencatat panggilan ke fungsi-fungsi yang berada di shared libraries yang digunakan oleh program, tujuannya agar menggunakan ltrace adalah untuk debugging logis, analisis perilaku program, dan rekayasa balik tingkat tinggi.

- Using GDB

```

nadiva_nathania@LAPTOP-JR1G744R:~$ nano buggy.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -g -o buggy buggy.c
nadiva_nathania@LAPTOP-JR1G744R:~$ gdb ./buggy
GNU gdb (Ubuntu 15.0.50.20240403-0ubuntu1) 15.0.50.20240403-git
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) break main
Breakpoint 1 at 0x1238: file buggy.c, line 18.
(gdb) run
Starting program: /home/nadiva_nathania/buggy

This GDB supports auto-downloading debuginfo from the following URLs:
<https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:18
18         buggy_function(5);
(gdb) next
array[0] = 0

```

```

Breakpoint 1, main () at buggy.c:18
18         buggy_function(5);
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
19         return 0;
(gdb) step
20     }
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 1617) exited normally]
(gdb) backtrace
No stack.
(gdb) quit

```

GDB ini untuk berinteraksi dengan program saat sedang berjalan atau untuk menganalisis keadaan program setelah terjadi crash. GDB dapat bekerja dengan berbagai bahasa pemrograman, termasuk C, C++, Ada, dan fortran. tujuan dari penggunaan GDB adalah debugging efisien dan analisis kode mendalam.

- **Kernel Debugging dengan dmesg**

Lihat kernel log

```
nadiva_nathan@LAPTOP-JR1G74HR:~$ dmesg
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 S
MP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug p
ty_legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000]   Intel GenuineIntel
[ 0.000000]   AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x0000000000f5ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x9a4e24, misc 0xe0bed7b2
[ 0.000000] Hyper-V: Nested features: 0x3e0101
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120
ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 2611.199 MHz processor
[ 0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.000000] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.000000] last_pfn = 0xf5600 max_arch_pfn = 0x40000000
[ 0.000000] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[ 0.000000] x86/PAT: Configuration [0-7]: WB WC UC- UC MB WP UC- WT
[ 0.001221] Using GB pages for direct mapping
[ 0.001289] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[ 0.001911] ACPI: Early table checksum verification disabled
[ 0.001955] ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
[ 0.001999] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.002065] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.002111] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
```

Follow kernel log

```
nadiva_nathan@LAPTOP-JR1G74HR:~$ dmesg -w
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 S
MP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=12 hv_utils.timesync_implicit=1 console=hvc0 debug p
ty_legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000]   Intel GenuineIntel
[ 0.000000]   AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x0000000000f5ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x9a4e24, misc 0xe0bed7b2
[ 0.000000] Hyper-V: Nested features: 0x3e0101
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120
ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 2611.199 MHz processor
[ 0.001533] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.001565] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.001599] last_pfn = 0xf5600 max_arch_pfn = 0x40000000
[ 0.001811] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[ 0.001833] x86/PAT: Configuration [0-7]: WB WC UC- UC MB WP UC- WT
[ 0.002323] Using GB pages for direct mapping
[ 0.003121] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[ 0.003155] ACPI: Early table checksum verification disabled
[ 0.003180] ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
[ 0.003221] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.003299] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.003344] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
```

Filter by level

```
nadiva_nathania@LAPTOP-JR1G744R:~$ dmesg -l err
[ 0.097299] PCI: Fatal: No config space access function found
[ 1.958854] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 1.961745] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.962976] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.963746] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 1.964325] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 6.284810] WSL (235) ERROR: CheckConnection: getaddrinfo() failed: -5
nadiva_nathania@LAPTOP-JR1G744R:~$ dmesg -l warn
[ 0.071599] Hyper-V: Disabling IBT because of Hyper-V bug
[ 0.132469] PCI: System does not support PCI
[ 0.246205] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
[ 1.834491] pulseaudio[272]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 2.205594] systemd-journald[42]: File /var/log/journal/8228cb8774e544abb6d2cf183e54a713/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 6.702477] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
nadiva_nathania@LAPTOP-JR1G744R:~$
```

Kernel debugging dengan dmesg adalah praktik yang menggunakan perintah untuk melihat log buffer pesan yang dihasilkan oleh kernel sistem operasi. Pesan-pesan ini mencakup informasi mengenai hardware saat booting, status driver, system call tertentu, dan terutama, laporan kesalahan atau peringatan yang terjadi pada tingkat kernel.

- **Memory Leak Detection dengan Valgrind**

Install valgrind

```
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo apt install valgrind
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
valgrind is already the newest version (1:3.22.0-0ubuntu3).
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
```

Buat program dengan memory leak

```
nadiva_nathania@LAPTOP-JR1G744R:~$ cat > memleak.c << 'EOF'
> #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
    return 0;
}
EOF
nadiva_nathania@LAPTOP-JR1G744R:~$ gcc -g -o memleak memleak.c
```

Jalankan dengan valgrind

```
nadiva_nathania@LAPTOP-JR1G744R:~$ valgrind --leak-check=full ./memleak
==1705== Memcheck, a memory error detector
==1705== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==1705== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==1705== Command: ./memleak
==1705==
==1705==
==1705== HEAP SUMMARY:
==1705==   in use at exit: 400 bytes in 1 blocks
==1705==   total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==1705==
==1705== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==1705==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==1705==    by 0x10915E: main (memleak.c:3)
==1705==
==1705== LEAK SUMMARY:
==1705==   definitely lost: 400 bytes in 1 blocks
==1705==   indirectly lost: 0 bytes in 0 blocks
==1705==   possibly lost: 0 bytes in 0 blocks
==1705==   still reachable: 0 bytes in 0 blocks
==1705==   suppressed: 0 bytes in 0 blocks
==1705==
==1705== For lists of detected and suppressed errors, rerun with: -s
==1705== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
```


Memory leak detection dengan valgrind adalah praktik menggunakan alat debugging dan profiling yang disebut valgrind untuk mengidentifikasi dan melokalisasi kebocoran memori dalam program, tujuannya agar menggunakan valgrind untuk mendeteksi memory leak adalah untuk memastikan kualitas, stabilitas, dan keandalan perangkat lunak.

- **Performance Analysis**

```
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  hwdmdata libnumal libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic
The following NEW packages will be installed:
  hwdmdata libnumal libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic linux-tools-common linux-tools-generic
0 upgraded, 7 newly installed, 0 to remove and 48 not upgraded.
Need to get 6303 kB of archives.
After this operation, 20.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libnumal amd64 2.0.18-1build1 [23.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libtraceevent1 amd64 1:1.8.2-1ubuntu2.1 [55.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 hwdmdata all 0.379-1 [29.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-common all 6.8.0-85.85 [762 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85 amd64 6.8.0-85.85 [5421 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-85-generic amd64 6.8.0-85.85 [1804 B]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-generic amd64 6.8.0-85.85 [10.8 kB]
Fetched 6303 kB in 6s (1120 kB/s)
Selecting previously unselected package libnumal:amd64.
(Reading database ... 50964 files and directories currently installed.)
Preparing to unpack .../0-libnumal_2.0.18-1build1_amd64.deb ...
Unpacking libnumal:amd64 (2.0.18-1build1) ...
Selecting previously unselected package libtraceevent1:amd64.
Preparing to unpack .../1-libtraceevent1_1%3a1.8.2-1ubuntu2.1_amd64.deb ...
Unpacking libtraceevent1:amd64 (1:1.8.2-1ubuntu2.1) ...
Selecting previously unselected package hwdmdata.
Preparing to unpack .../2-hwdmdata_0.379-1_all.deb ...
Unpacking hwdmdata (0.379-1) ...
Selecting previously unselected package linux-tools-common.
Preparing to unpack .../3-linux-tools-common_6.8.0-85.85_all.deb ...
Unpacking linux-tools-common (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85.
Preparing to unpack .../4-linux-tools-6.8.0-85_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85 (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-6.8.0-85-generic.
Preparing to unpack .../5-linux-tools-6.8.0-85-generic_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Selecting previously unselected package linux-tools-generic.
Preparing to unpack .../6-linux-tools-generic_6.8.0-85.85_amd64.deb ...
Unpacking linux-tools-generic (6.8.0-85.85) ...
Setting up hwdmdata (0.379-1) ...
Setting up libnumal:amd64 (2.0.18-1build1) ...
Setting up libtraceevent1:amd64 (1:1.8.2-1ubuntu2.1) ...
Setting up linux-tools-common (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85 (6.8.0-85.85) ...
Setting up linux-tools-6.8.0-85-generic (6.8.0-85.85) ...
Setting up linux-tools-generic (6.8.0-85.85) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
nadiva_nathania@LAPTOP-JR1G744R:~$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
```

Performance analysis adalah proses pengumpulan, pemeriksaan, dan evaluasi data tentang bagaimana sebuah sistem komputer, perangkat lunak, atau komponen tertentu beroperasi dan menggunakan sumber daya selama eksekusi, tujuannya agar melakukan analisis

kinerja adalah untuk optimasi dan troubleshooting demi memastikan sistem beroperasi secara efisien dan andal.